

Fall 25, 4100 Problem Set 7

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Let $A \in \mathbb{C}^{n \times n}$ and W be a subspace of $\mathbb{C}^n$. Recall that W is an A -invariant subspace if and only if for all $\mathbf{x} \in W$, $A\mathbf{x} \in W$. Prove that the generalized eigenspace, $\text{null}(A - \lambda I)^n$ is A -invariant for any eigenvalue λ and n .
2. For each of the following matrices find the eigenvalues, eigenvectors, and generalized eigenvectors, and state the Jordan form

a. $\begin{bmatrix} -3 & -1 \\ 1 & -5 \end{bmatrix}$

b. $\begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & -5 \\ 0 & 0 & 3 \end{bmatrix}$

3. a. Given $J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$, find a formula for the matrix exponential e^{Jt} .
b. Use the result above and 2a. to solve the linear system

$$\begin{aligned} x_1' &= -3x_1 - x_2 \\ x_2' &= x_1 - 5x_2 \end{aligned}$$

with initial values $x_1(0) = 1$, $x_2(0) = 2$

4. Let A be such that $p_A(\lambda) = |A - \lambda I| = (\lambda - 4)(\lambda - 2)^5(\lambda + 1)^2$ with $\text{nullity}(A - 4I) = 1$, $\text{nullity}(A - 2I) = 3$, and $\text{nullity}(A + I) = 1$. Find all possible Jordan forms of A up to reordering.
5. Let A be such that $p_A(\lambda) = |A - \lambda I| = (\lambda - 3)^7(\lambda + 2)^6$ with minimal polynomial, $\mu_A(x) = (x - 3)^5(x + 2)^3$ and geometric multiplicities, $\text{nullity}(A - 3I) = 2$ and $\text{nullity}(A + 2I) = 3$. Find the **unique** (up to reordering) Jordan form of A .
6. Let $A \in \mathbb{R}^{n \times n}$ be such that $A^k = I$. Prove that A is diagonalizable. (Hint: a polynomial p has a simple zero at z_0 iff $p(z_0) = 0$ but $p'(z_0) \neq 0$.)